UNIT 4 · TOPIC 2(A)

Engine Basics and the Valve System

Syllabus topic: Various systems of IC engines: operation, adjustment and troubleshooting

Terminology

Bore D, stroke L, top and bottom dead centres (TDC/BDC), clearance volume Vc.

$$\text{Swept volume } V_s = (\pi/4)D^2L \text{ per cylinder; compression ratio } r = (V_s + V_c)/V_c$$

Example. D = 100 mm, L = 120 mm → Vs = **0.942 L/cyl** (4-cyl = 3.77 L engine). With Vc = 55 cm³ → r = (942 + 55)/55 ≈ **18.1** — a typical DI diesel.

Four-stroke and two-stroke cycles

1. Suction.
2. Compression.
3. Power (expansion).
4. Exhaust.

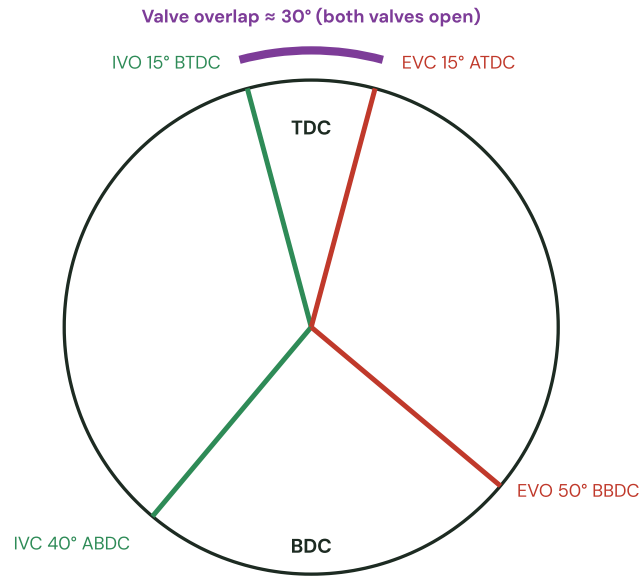
- **Four-stroke:** one power stroke per 2 crank revolutions — the camshaft runs at HALF crank speed.
- **Two-stroke:** a power stroke every revolution, using ports + scavenging. Used in light equipment (older sprayer engines, chainsaws).

Valve mechanism and timing

The overhead-valve (I-head) layout is standard on tractor diesels. Drive chain:

1. Cam
2. Tappet (follower)
3. Pushrod
4. Rocker arm
5. Valve, closed against its spring

Timing gears or a chain drive the camshaft at half crank speed.



Typical diesel valve timing (indicative values; redrawn from the source figure)

Valves do NOT open or close exactly at the dead centres:

Event	Typical timing	Reason
Intake opens	~10–25° BTDC	—
Intake closes	~25–45° ABDC	Ram effect fills the cylinder
Exhaust opens	~30–50° BBDC	Blow-down
Exhaust closes	~10–25° ATDC	—
VALVE OVERLAP	~20–40° around TDC (both open)	Scavenges residual gas

Valve (tappet) clearance — the classic adjustment

1. Set the clearance cold, with a feeler gauge.
2. Typical values: ~**0.25 mm intake / 0.30 mm exhaust** — always follow the maker's specification and setting sequence.

Clearance	Effect
TOO SMALL	Valve held slightly open when hot → leakage, loss of compression, BURNT valves
TOO LARGE	Noisy tappets; late opening / early closing → poor breathing, reduced power

- The **decompression lever** holds the valves open for easy hand cranking.

Practice MCQs

Q36. In a 4-stroke engine the camshaft rotates at:

(a) Crank speed
(b) Half crank speed
(c) Twice crank speed
(d) Quarter speed

Q37. A 4-stroke engine gives one power stroke per:

(a) 1 revolution
(b) 2 revolutions
(c) 4 revolutions
(d) ½ revolution

Q38. Bore 100 mm, stroke 120 mm → swept volume per cylinder ≈

(a) 0.55 L
(b) 0.79 L
(c) 0.94 L
(d) 1.2 L

Q39. With Vc = 55 cm³ and Vs = 942 cm³, compression ratio ≈

(a) 14
(b) 16
(c) 18
(d) 22

Q40. Valve overlap is the period when:

(a) Both valves are closed
(b) Both intake and exhaust are open near TDC
(c) Piston stops
(d) Injection occurs

Q41. The exhaust valve opens:

(a) At BDC exactly
(b) 30–50° BEFORE BDC
(c) At TDC
(d) After TDC only

Q42. Too SMALL a valve (tappet) clearance causes:

(a) Noise only
(b) Valve leakage and burnt valves
(c) Higher power
(d) Better idle

Q43. Too LARGE a clearance causes:

(a) Burnt valves
(b) Noisy operation with late opening and reduced breathing
(c) Overheating
(d) Oil dilution

Q44. Valve clearance is measured with a:

(a) Micrometer
(b) Feeler gauge
(c) Dial bore gauge
(d) Vernier

Q45. The decompression lever:

(a) Increases r
(b) Holds valves open to ease cranking
(c) Stops fuel
(d) Adjusts governor

Answers

Q36 — (b) Half crank speed

Q37 — (b) 2 revolutions

Q38 — (c) 0.94 L — *Vs = (π/4)(1.0)²(1.2) dm³ = 0.942 L.*

Q39 — (c) 18 — *r = (942 + 55)/55 = 18.1 — a typical DI diesel.*

Q40 — (b) Both intake and exhaust are open near TDC

Q41 — (b) 30–50° BEFORE BDC

Q42 — (b) Valve leakage and burnt valves — *Clearance compensates thermal expansion; burnt valves (too small) vs noisy tappets (too large) is a perennial exam pair.*

Q43 — (b) Noisy operation with late opening and reduced breathing

Q44 — (b) Feeler gauge

Q45 — (b) Holds valves open to ease cranking

Sources: Liljedahl et al. — Tractors and their Power Units; Jagdishwar Sahay; manufacturer service practice; TREM notifications; NPTEL farm power lectures.